Compute Dot Product

$\mathbf{v} = \langle a_1, b_1 \rangle \quad \mathbf{w} = \langle a_2, b_2 \rangle, \quad \mathbf{v} \cdot \mathbf{w} = a_1 a_2 + b_1 b_2.$

Definition of Dot Product If $\mathbf{v} = \langle a_1, b_1 \rangle$ and $\mathbf{w} = \langle a_2, b_2 \rangle$, the dot product is defined as $\mathbf{v} \cdot \mathbf{w} = a_1 a_2 + b_1 b_2$.

Note that the dot product results in a *scalar*, not a vector.

$\mathbf{v} = \langle -2, 2 \rangle \quad \mathbf{w} = \langle 3, 14 \rangle,$

1. Given $\mathbf{v} = \langle -2, 2 \rangle$ and $\mathbf{w} = \langle 3, 14 \rangle$, find

a. $\mathbf{v} \cdot \mathbf{w}$

$= (-2)(3) + (2)(14)$

$= 22$

b. $\mathbf{v} \cdot \mathbf{v}$

$= (-2)(-2) + (2)(2)$

$= 8$

c. $\mathbf{w} \cdot \mathbf{w}$

$= 3^2 + 14^2$

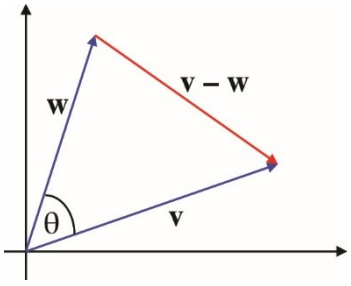
$= 205$

Properties of Dot Product If **u**, **v**, and **w** are vectors, and c is a scalar, then

1. $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$
2. $(c\mathbf{v}) \cdot \mathbf{w} = c(\mathbf{v} \cdot \mathbf{w})$
3. $\mathbf{v} \cdot (\mathbf{w} + \mathbf{u}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{u}$
4. $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$
5. $\mathbf{0} \cdot \mathbf{v} = 0$
6. $c(\mathbf{v} \cdot \mathbf{w}) = (c\mathbf{v}) \cdot \mathbf{w} = \mathbf{v} \cdot (c\mathbf{w})$

Find the Angle Between Two Vectors

Given two vectors **v** and **w** in standard position, the angle θ between **v** and **w** is the angle formed by the two line segments representing **v** and **w** where $0 \le \theta \le 180°$.



Angle Between Two Vectors

If θ is the angle between two nonzero vectors **v** and **w**, then

$$\cos\theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} \quad \text{and} \quad \theta = \cos^{-1}\left(\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|}\right).$$

---

2. Find the angle between $\mathbf{v} = -2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{w} = 3\mathbf{i} + 4\mathbf{j}$. Round to the nearest tenth of a degree.

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

$\|\mathbf{v}\| = \sqrt{\mathbf{v}\cdot\mathbf{v}} = \sqrt{(-2)^2 + 3^2} = \sqrt{13}$ $\qquad$ $\mathbf{v}\cdot\mathbf{w} = (-2)(3) + (3)(4) = 6$

$\|\mathbf{w}\| = \sqrt{\mathbf{w}\cdot\mathbf{w}} = \sqrt{3^2 + 4^2} = 5$

$$\cos\theta = \frac{6}{5\sqrt{13}}$$

$$\theta = \cos^{-1}\frac{6}{5\sqrt{13}} \approx 70.6°$$

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

$\cos\theta = \dfrac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|}$ may also be written as $\mathbf{v}\cdot\mathbf{w} = \|\mathbf{v}\|\|\mathbf{w}\|\cos\theta.$

If $\mathbf{v}\cdot\mathbf{w} > 0$, then $\cos\theta > 0 \Rightarrow 0° \le \theta < 90°.$ If $\mathbf{v}\cdot\mathbf{w} < 0$, then $\cos\theta < 0 \Rightarrow 90° < \theta \le 180°.$

If $\mathbf{v}\cdot\mathbf{w} = 0$, then $\cos\theta = 0 \Rightarrow \theta = 90°$ and we say that **v** and **w** are orthogonal.
Orthogonal vectors drawn "tail-to-tail" will form a right angle.

$$\mathbf{v}\cdot\mathbf{w} = 0.$$

Orthogonal VectorsTwo vectors **v** and **w** are orthogonal if and only if

If **v** and **w** are parallel, then $\mathbf{v}\cdot\mathbf{w} = \|\mathbf{v}\|\|\mathbf{w}\|$ or $\mathbf{v}\cdot\mathbf{w} = -\|\mathbf{v}\|\|\mathbf{w}\|$.

Determine if the given vectors are orthogonal, parallel, or neither.

3. $\mathbf{k} = \langle 1.8, 0.3\rangle,\ \mathbf{s} = \langle -9, -1.5\rangle$

$\mathbf{k}\cdot\mathbf{s} = (1.8)(-9) + (0.3)(-1.5) = -16.65$

$\|\mathbf{k}\| = \sqrt{1.8^2 + 0.3^2} = 0.3\sqrt{37}$ $\qquad \|\mathbf{k}\|\|\mathbf{s}\| = 0.3\sqrt{37}\cdot 1.5\sqrt{37} = 16.65$

$\|\mathbf{s}\| = \sqrt{(-9)^2 + (-1.5)^2} = 1.5\sqrt{37}$ $\qquad$ parallel

4. $\mathbf{b} = \dfrac{2}{3}\mathbf{i} - 12\mathbf{j},\ \mathbf{a} = 15\mathbf{i} + \dfrac{5}{6}\mathbf{j}$

$\mathbf{b}\cdot\mathbf{a} = \left(\dfrac{2}{3}\right)(15) + (-12)\left(\dfrac{5}{6}\right) = 0$

orthogonal

5. $\mathbf{r} = \langle -6, 15\rangle,\ \mathbf{s} = \langle 5, -2\rangle$

$\mathbf{r}\cdot\mathbf{s} = (-6)(5) + (15)(-2) = -60$

$\|\mathbf{r}\| = \sqrt{(-6)^2 + 5^2} = \sqrt{61}$

$\|\mathbf{s}\| = \sqrt{5^2 + (-2)^2} = \sqrt{29}$ $\qquad \|\mathbf{r}\|\|\mathbf{s}\| = \sqrt{61}\cdot\sqrt{29} \approx 42$ $\qquad$ Neither

---

Decompose Vectors into Two Orthogonal Vectors

Decomposition of a Vector into Orthogonal Vectors

If **v** and **w** are two nonzero vectors, then

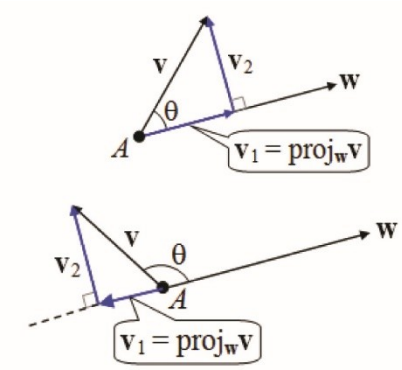
1. The vector projection of **v** onto **w** is given by

$$\text{proj}_{\mathbf{w}}\mathbf{v} = \frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{w}\|^2}\mathbf{w}.$$

2. To decompose **v** into vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ parallel to **w** and orthogonal to **w**:

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

$$\mathbf{v}_1 = \text{proj}_{\mathbf{w}}\mathbf{v} = \frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{w}\|^2}\mathbf{w} \quad \text{and} \quad \mathbf{v}_2 = \mathbf{v} - \mathbf{v}_1$$



6. Given $\mathbf{v} = 3\mathbf{i} - \mathbf{j}$ and $\mathbf{w} = -8\mathbf{i} - 3\mathbf{j}$,

$$\frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{w}\|^2} = \frac{(3)(-8)+(-1)(-3)}{(-8)^2+(-3)^2} = -\frac{21}{73}$$

a. Find $\text{proj}_{\mathbf{w}}\mathbf{v}$.

$$= -\frac{21}{73}(-8\mathbf{i} - 3\mathbf{j})$$

$$= \frac{168}{73}\mathbf{i} + \frac{63}{73}\mathbf{j}$$

b. Find vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ such that $\mathbf{v}_1$ is parallel to $\mathbf{w}$, $\mathbf{v}_2$ is orthogonal to $\mathbf{w}$, and $\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{v}$.

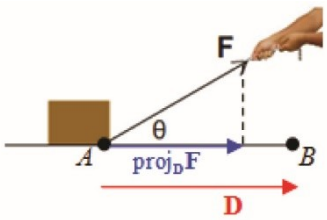
$$\mathbf{v}_1 = \text{proj}_{\mathbf{w}}\mathbf{v}_1 = \left\langle \frac{168}{73}, \frac{63}{73} \right\rangle$$

$$\mathbf{v}_2 = \mathbf{v} - \mathbf{v}_1 = \langle 3, -1 \rangle - \left\langle \frac{168}{73}, \frac{63}{73} \right\rangle = \left\langle \frac{51}{73}, -\frac{136}{73} \right\rangle$$

Use Vectors to Compute Work

Definition of Work

If $\mathbf{D}$ is the displacement vector of an object in moving the object in a straight line from points A to B under a constant force $\mathbf{F}$, then the work W done is computed by



1. $W = \|\text{proj}_{\mathbf{D}}\mathbf{F}\|\|\mathbf{D}\| \quad W = \|\mathbf{F}\|\|\mathbf{D}\|\cos\theta,$ or

where θ is the angle between $\mathbf{F}$ and $\mathbf{D}$.

2. $W = \mathbf{F}\cdot\mathbf{D}$ note: work is not a vector

---

7. A force of 85 N is applied to move an object 20 m horizontally. If the force is directed 44° upward from the horizontal, find the work done in moving the object.

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

$\underset{\rightarrow}{F} = 85\ \text{N}$

$\underset{\rightarrow}{D} = 20\mathbf{i}$

$\theta = 44°$

$$W = \left\|\underset{\rightarrow}{F}\right\|\left\|\underset{\rightarrow}{D}\right\| \cos\theta$$

$$= (85\ \text{N})(20\ \text{m}) \cos 44° \approx 1223\ \text{N} \cdot \text{m}$$

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.